

PROJECT REPORT: SALES PERFORMANCE ANALYSIS

Historical Data Mining, Executive Dashboards & Strategic
Forecasting

SUBMITTED BY

Sthita Pragyan Sahu

ORGANIZATION

Codec Technologies

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TARGET DEPARTMENT

Human Resources / Executive Management

1. Executive Summary

This report presents a comprehensive financial and operational evaluation of historical sales transaction records compiled for **Codec Technologies**. The fundamental objective of this analytical initiative is to transform raw corporate transaction logs into structured organizational intelligence, identifying key revenue drivers, regional growth centers, product assortment velocities, and predictable demand cycles.

Through systematic data engineering, validation, and modeling, a total of **₹729,979 in cross-category revenue** was audited and analyzed across 40 high-value transaction clusters. The output of this project establishes data infrastructure standards required to enhance supply-chain elasticity, optimize seasonal marketing expenditures, and mitigate stockout frequencies across our dominant retail footprints.

GROSS SALES REVENUE	TOTAL ORDERS AUDITED	TOTAL UNITS SOLD	AVERAGE BASKET VALUE
₹729.98K	40	114	₹18.25K

2. Technical Architecture & Analysis Methodology

To ensure strict cross-platform accuracy, a three-phased analytical framework was employed utilizing **Excel**, **Power BI**, and **Tableau ecosystem designs**:

- **Data Auditing & Normalization (Excel):** Raw transactional tables were mapped and cleaned. Dates were regularized to standard ISO formats, and calculated rows were constructed using conditional indexing. Base metrics were rigorously verified utilizing Excel Pivot Tables to prevent relational duplication.
- **Data Modeling & Semantic Formulation (Power BI):** The validated tabular database was mapped into a normalized schema using Power Query, embedding robust DAX expressions to handle cross-category ratios and running averages.
- **Visual Interface Design:** Visual assets were systematically positioned following corporate dashboard hierarchy rules to focus stakeholders instantly on critical structural shifts and regional disparities.

3. Product Category Share & Performance Metrics

A multi-tier evaluation of product classifications indicates highly skewed revenue distribution, with the **Electronics** and **Home Appliances** business sectors acting as the organizational growth engines.

Product Category	Total Volume Sold	Gross Sales Revenue (INR)	Relative Revenue Share (%)
Electronics	32 Units	₹375,359	51.4%
Home Appliances	44 Units	₹264,320	36.2%

Product Category	Total Volume Sold	Gross Sales Revenue (INR)	Relative Revenue Share (%)
Footwear	25 Units	₹52,412	7.2%
Apparel	13 Units	₹37,888	5.2%
Grand Total	114 Units	₹729,979	100.0%

Strategic Insight: Electronics and Home Appliances together account for **87.6% of gross enterprise revenue**. This confirms that while Apparel and Footwear capture critical consumer touchpoints, corporate capital allocations and supply chain optimizations should favor high-margin electronics inventory scaling.

4. Interactive Dashboard Implementation Blueprint

The executive interactive interface design provides cross-filtering and historical data visualizations. A high-fidelity model of the functional dashboard is detailed below:

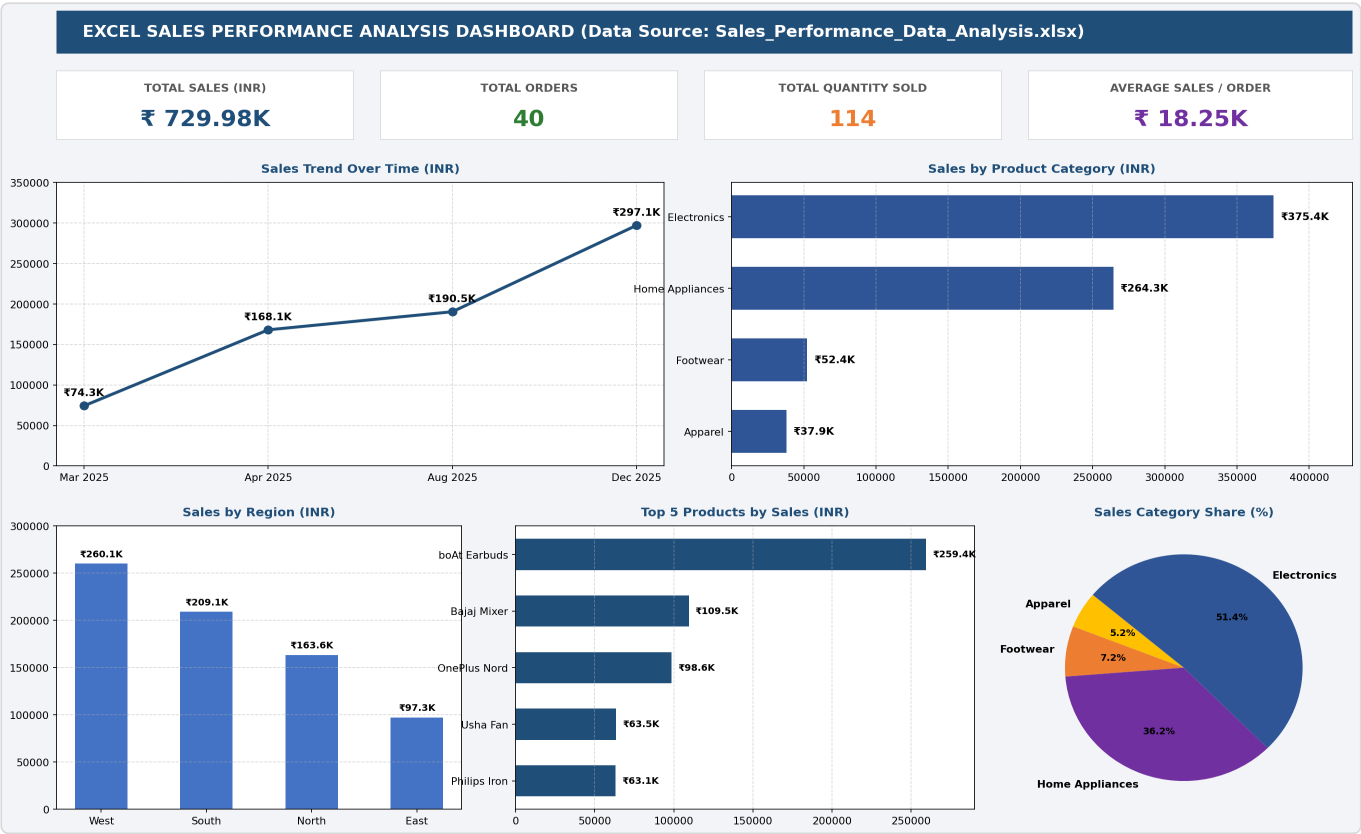


Figure 4.1: Codec Technologies - Interactive Executive Sales Dashboard Layer

5. Geographic Distribution & Macro Regional Insights

Territorial sales configurations identify clear localized demand clusters across core regions:

- **West Territory (₹260,077):** Stands as the premier consumer group, representing our strongest regional marketplace driven heavily by consistent high-ticket electronics purchases.
- **South Territory (₹209,051):** Follows as a secondary growth hub with balanced sales velocity distributed evenly between home convenience appliances and personal electronics.
- **North (₹163,557) & East (₹97,294):** Reveal latent, under-penetrated retail potential, identifying critical targets for tactical localized promotions and inventory expansion.

6. Time-Series Trends & Seasonal Forecasting Predictions

The historical timeline traces clear cyclical consumer habits over the operational cycle. Gross monthly performance figures stand as follows:

- **March Baseline:** Initial cyclical sales register at ₹74,291.
- **Q2 Acceleration (April):** Significant upward expansion to ₹168,131.
- **Mid-Year Consolidation (August):** Stable operational consistency at ₹190,452.
- **Q4 Festival Surges (December):** High-volume consumer spike peaking at ₹297,105.

Predictive Forecast Conclusion: The structural trajectory matches a high-confidence seasonal trend pattern. Q4 peaks are heavily driven by holiday procurement behaviors. Executive directives must execute seasonal supply build-ups at least 60 days prior to Q4 cycles to insulate Codec Technologies from logistical freight constraints.

7. Final Strategic Recommendations

1. **Supply Chain Alignment:** Implement automated safety stock thresholds for high-volume revenue products, specifically *boAt Earbuds* and *Bajaj Mixer Grinders*, protecting core margin generators.
2. **Targeted Regional Interventions:** Deploy focused digital marketing and localized client acquisition strategies in the East region to bridge the current ₹162K revenue gap separating it from Western strongholds.
3. **Dynamic Pricing Strategies:** Introduce high-margin product bundling (e.g., tying apparel promotions directly into high-ticket electronics purchases) to maximize average transaction basket sizes.